

Mathematics

A Cultural Heritage

Lecture 1

**Introduction:
Real-time discussion**

Lecture Outline

1. Course learning outcomes
2. Teaching staff
3. Assessment and student work
4. Nature of the course
5. Level of Mathematics required
6. Classwork
7. Reading assignment for this week

Course Learning Outcomes

1. Appreciate and describe the beauty, the utility, and the “way” of mathematics.
2. Comprehend and describe how mathematics was and is being developed as a work of human culture.
3. Investigate and describe the interplay between mathematics and other areas of human culture.
4. Investigate and explain the role of mathematics in the development of civilization.

Course Website

1. HKU Portal → My eLearning → Moodle courses
(<https://moodle.hku.hk/course/view.php?id=127378>)

The screenshot displays the HKU Moodle interface for the course "CCST5037 Mathematics: A Cultural Heritage (Online) [Section 1A, 2024]". The top navigation bar includes the HKU Moodle logo, "Dashboard", "My courses", and user information for Benjamin Robert Kane (BK) with an "Edit mode" toggle.

Left Sidebar:

- Course Information
 - News announcement
 - Default course forum
 - News announcement
 - Default course forum
 - Course Information
 - Forum for forming groups o...
 - Discussion about the videos
 - How to register for the Tuto...
- Zoom Lecture Links
- Zoom Tutorial Links
 - Tutorial Registration
- Tutorial Materials
 - Tutorial 1 slides
 - Tutorial 1 worksheet
 - Tutorial 2 slides
 - Read before Tutorial 3 (Deb...
 - Read before Tutorial 3 (Deb...
 - Tutorial 4 slides

Main Content Area:

The main area features a course banner with the title "CCST5037 Mathematics: A Cultural Heritage (Online) [Section 1A, 2024]". Below the banner are tabs for "Course", "Settings", "Participants", "Grades", "Reports", and "More". The "Course" tab is active, showing a "Course Information" section with a "Collapse all" link. Below this are several items, each with a red speech bubble icon:

- News announcement
- Default course forum
- News announcement
- Default course forum
- Course Information (with a globe icon)
- Forum for forming groups or general discussion

Right Sidebar:

The right sidebar contains two main sections:

- Lecture recording via Zoom:** Includes a "Create Zoom Meeting" button and a note: "Please note recording will start automatically when Zoom starts." Below this is a green "Create the meeting" button.
- Administration:** A list of administrative tasks under "Course administration":
 - Settings
 - Course completion
 - Users
 - Filters
 - Reports
 - Gradebook setup
 - Outcomes
 - Badges
 - Import
 - Backup
 - Restore
 - Reset
 - Question bank

Assessment

Task	Weighting
In-class worksheets	15%
Tutorial participation	25%
Mathematical Articles Assignment	15%
Group Essay Plan	10%
Group Presentation	15%
Group Essay	20%

Other obligations

1. Reading assignments
2. Watching videos
3. Viewing/interacting with website content

Analyzing all of the above in the context of this course.

Tutorials

1. Attendance **Compulsory**

2. Small class size

3. Objective and focus:

- Discussion / Interaction
- Working on problems
- Presenting your view/answers/solutions
- Raising questions

Tutorial logistics

1. Tutorials start on Tuesday, **Sept. 15, 2026.**
2. **No** tutorials during the reading week.
3. Tutorial Registration: From **Sept. 2, 2026, 15:00** to **Sept. 13, 2026, 12:00 (Sunday):**

<https://moodle.hku.hk/mod/choicegroup/view.php?id=3783134>

Link on Moodle page

Written Assignments

1. Two written assignments
2. Work on your own
3. Typical length 300-500 words
4. Expect **at least** 1.5 weeks to complete
5. Comparison and contrast to reading assignments part of the basis of assignment

Group Essay / Presentation

1. Groups of size 2-3 (subject to enrollment sizes)
2. 2000 – 3000 words
3. On book(s) (or other media, e.g. movie) about mathematics that is course-relevant
4. If multiple books chosen, essay must relate them
5. Project plan due by **October 16, 2026, 16:55**
6. Oral presentations (short) in last 2 tutorials
7. Essay due by **December 1, 2026, 16:55.**

Grade determination

Grade	Requirement
A	Demonstrate thorough knowledge and skills required for attaining all the course learning outcomes. Show strong analytical and critical abilities and logical thinking. Apply highly effective organizational and presentational skills.
B	Demonstrate substantial command of knowledge and skills required for attaining at least most of the course learning outcomes. Show evidence of analytical and critical abilities and logical thinking. Apply effective organizational and presentational skills.
C	Demonstrate general but incomplete command of knowledge and skills required for attaining most of the course learning outcomes. Show evidence of some analytical and critical abilities and logical thinking. Apply moderately effective organizational and presentational skills.
D	Demonstrate partial but limited command of knowledge and skills required for attaining some of the course learning outcomes. Show evidence of some coherent and logical thinking, but with limited analytical and critical abilities. Apply limited or barely effective organizational and presentational skills.
F	Demonstrate little or no evidence of command of knowledge and skills required for attaining the course learning outcomes. Lack of analytical and critical abilities, logical and coherent thinking. Organization and presentation skills are minimally effective or ineffective.

Acknowledgements

1. This Common Core course is based on a previous course, with the same course title, developed and taught by Professor Man-keung Siu (蕭文強教授).
2. Dr. Nam-Kiu Tsing was one of the originators of this incarnation of the course.
3. Dr. Ka Wai Rachel Lui was one of the originators of this incarnation of the course and the main creator of the tutorial materials.

Caveats

1. This is not a “regular” mathematics course (i.e., Calculus, Linear Algebra, etc.)
2. The goal and focus is not solving numerical problems or proving theorems
3. The main work involves **reading, writing, analyzing, discussing, and debating** about how mathematics plays a role in human society and culture and **presenting** own's views.

Assignment 1 (Project Plan) Logistics

- Choose a tentative title and explain your choice
- Include names and UIDs
- Find a primary source and other basic references
- Determine a central theme and show how it fits into the course
- Connect to **Course Learning Outcomes**
- 200-400 Words
- Brief description of proposed essay content
- Watch out for **copying/plagiarism**
- Due date: **Friday, 16th October, 2026, 16:55**

Course Learning Outcomes

1. Appreciate and describe the beauty, the utility, and the “way” of mathematics.
2. Comprehend and describe how mathematics was and is being developed as a work of human culture.
3. Investigate and describe the interplay between mathematics and other areas of human culture.
4. Investigate and explain the role of mathematics in the development of civilization.

Assignment 2 Logistics

- Start collecting clippings that have to do with mathematics from current (published on or after 19th September, 2026) **newspapers, magazines, or webpages** (in English or in Chinese, whose target audience is the general public).
- Do **not** take an excerpt out of a book or a magazine or a website which is **specifically on mathematics or science**.
- Your source should **not** be media explicitly designed for experts to write in, even if written for laypeople.
- As a **general guideline**, ask whether most people would see this media in their **daily lives** or only in a **specialized search**.
- Connection to math **explicitly** mentioned in article
- Target number of items is 10.

Assignment 2 Logistics

- Record **source** and **date**, and write a paragraph or so (in English) **explaining** why you think that this article **fits the selection criteria**, how it connects to the **Learning Outcomes** of this course, and **what led you to choose it**. **Persuasion skills** will also be assessed when reading your descriptions.
- Provide a description/translation if the article is in Chinese.
- Do **not** simply give articles containing **numbers or data!**
- Submit **type-written** passages and front page of 10 items in single **PDF** file (through Moodle).
- Due date: **Monday, 30th November, 2026, 16:55.**

Examples

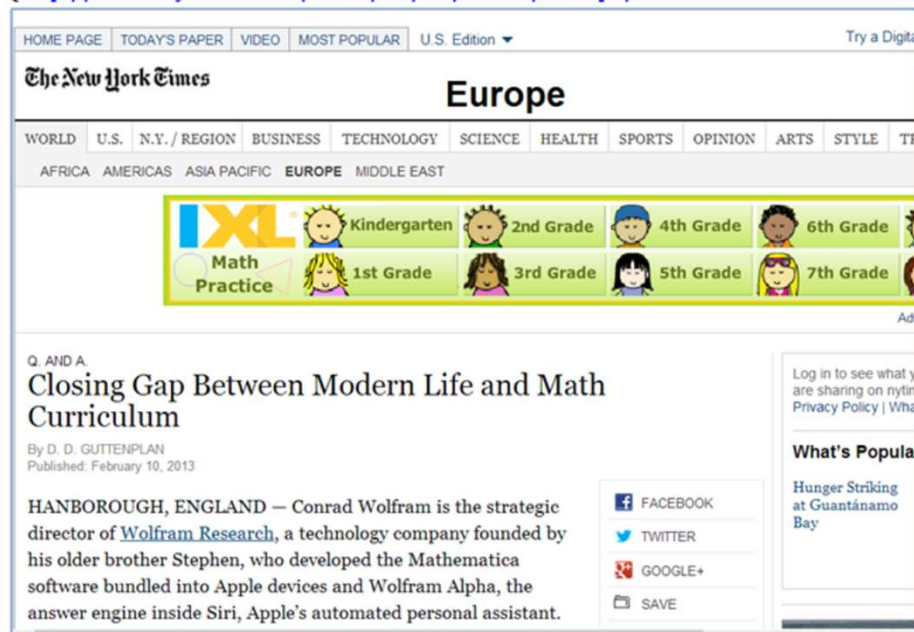
title

date

The New York Times: "Closing Gap Between Modern Life and Math Curriculum", 10th Feb 2013.

(<http://www.nytimes.com/2013/02/11/world/europe/11iht-educside11.html?ref=mathematics>)

source

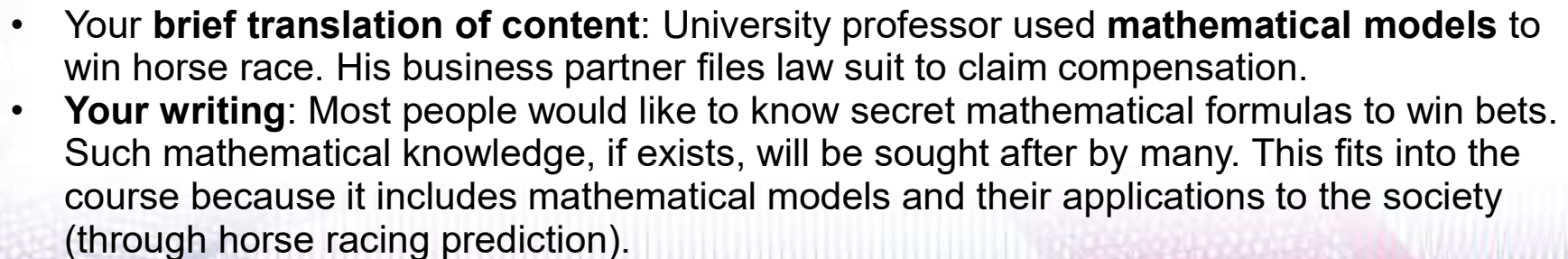


clipping

Conrad Wolfram, strategic director of Wolfram Research, is campaigning to change the mathematical curriculum in schools by making it more applicable to “real life”. He argues that the mathematics education forced on children today is not at all useful in daily activities, and that modern life has become so quantitative (eg. through the use of statistics in medicine, comparing mortgages and understanding financial crises) we must completely change the mathematical curriculum so that we can deal with these new problems. He believes we should be retrained in computer programming, mental arithmetic and complex mathematics using technology.

Your writing

Example of raw material in Chinese (must summarize in English)



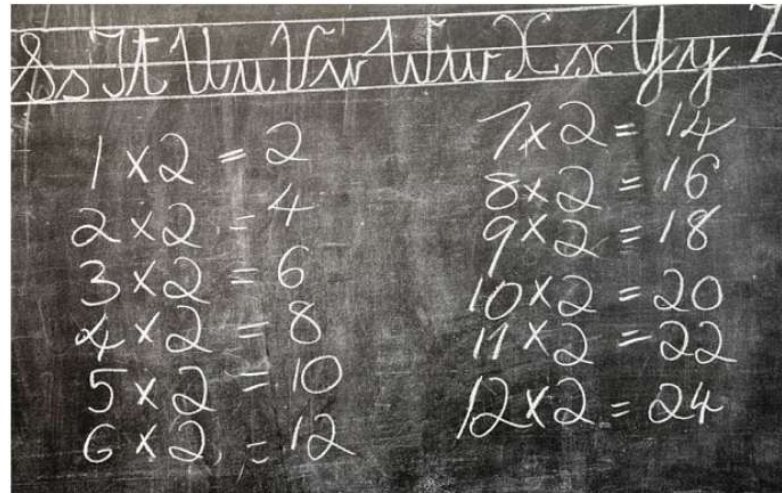
Other Examples

The Telegraph

<http://www.telegraph.co.uk/education/educationnews/11031288/Britains-maths-policy-simply-doesnt-add-up.html>

Britain's maths policy simply doesn't add up

As a woman wins the greatest prize, it's time to liberate girls from our test-driven culture



For the first time, the Fields Medal has been awarded to a woman – Maryam Mirzakhani, who is responsible for what fellow mathematicians call “probably the theorem of the decade” Photo: Alamy

By Jo Boaler

6:10AM BST 14 Aug 2014

Yesterday was an important day for gender equity in mathematics. For the first time, the Fields Medal was awarded to a woman – my colleague at Stanford University, Maryam Mirzakhani. It is a much-deserved prize for one of the most exciting mathematicians of our time.

Examples that are NOT acceptable

- Published on or before Sept. 18, 2026
- Not published in **mass media** with **general public** as audience
 - E.g., Journal articles like Nature, Bulletin of the American Mathematical Society, personal blogs, etc.
- Articles containing data but with **no analysis**.
 - Example: Bad: An article about the “Million People March against Police Brutality” – Description “One Million is a number.”
 - Better: An article about “how big” one million is; how long would it take to count to one million seconds?

Examples that are NOT acceptable

Example: <http://nytimes.stats.com/nba/scoreboard.asp>

HOME PAGE TODAY'S PAPER VIDEO MOST POPULAR TIMES TOPICS										Log In Register Now	
The New York Times Tuesday, January 19, 2016										Search All NYTimes.com Go	
WORLD U.S. N.Y. / REGION BUSINESS TECHNOLOGY SCIENCE HEALTH SPORTS OPINION ARTS STYLE TRAVEL JOBS REAL ESTATE AUTOS											
BASEBALL PRO FOOTBALL COLLEGE FOOTBALL PRO BASKETBALL COLLEGE BASKETBALL HOCKEY SOCCER GOLF TENNIS GLOBAL SPORTS											
Scoreboard Stats Standings Teams Players Player News Injuries Transactions										LATEST SCORES AND RESULTS	
NBA Scoreboard										• M.L.B. N.B.A. N.F.L. N.H.L.	
Previous Day Monday, January 18 Next Day										• NCAA Basketball: Men's Women's	
Refresh: 60 Sec 120 Sec Off										• Soccer: M.L.S. E.P.L. Champions League	
										• Golf: P.G.A. L.P.G.A. Web.com Champions European	
										• Tennis: Men Women	
										• Motor Sports: NASCAR Xfinity Truck IndyCar Formula 1	
Final-OT - 2 OT 1 2 3 4 OT 2OT Tot											
Philadelphia 29 18 16 33 11 6 113											
New York 29 23 24 20 11 12 119											
Preview Box Recap											
Leaders Philadelphia New York											
Points J. Okafor 20 A. Afflalo 25											
Rebounds N. Noel 13 K. Porzingis 12											
Assists I. Smith 16 C. Anthony 7											
Final-OT - 2 OT 1 2 3 4 OT 2OT Tot											
Utah 28 15 21 31 13 11 119											
Charlotte 29 23 21 22 13 16 124											
Preview Box Recap											
Leaders Utah Charlotte											
Points G. Hayward 36 K. Walker 52											
Rebounds R. Gobert 14 2 tied with 9											
Assists G. Hayward 9 K. Walker 8											
Final 1 2 3 4 Tot											
Chicago 26 30 27 28 111											
Detroit 32 21 26 22 101											
Preview Box Recap											
Leaders Chicago Detroit											
Final 1 2 3 4 Tot											
Portland 32 27 35 14 108											
Washington 17 40 19 22 98											
Preview Box Recap											
Leaders Portland Washington											
Points C. McCollum 25 G. Temple 18											
Rebounds M. Plumlee 11 M. Gortat 13											
Assists M. Plumlee 7 J. Wall 10											
Final 1 2 3 4 Tot											
New Orleans 26 25 25 23 99											
Memphis 24 25 23 29 101											
Preview Box Recap											
Leaders New Orleans Memphis											
Points J. Holiday 23 4 tied with 16											
Rebounds A. Davis 8 M. Gasol 10											
Assists J. Holiday 9 M. Conley 10											
Final 1 2 3 4 Tot											
Orlando 17 21 21 22 81											
Atlanta 24 27 21 26 98											
Preview Box Recap											
Leaders Orlando Atlanta											

Group Project Logistics

1. Work in Groups of **3** (same tutorial group) – subject to confirmation of class size
2. 2000 – 3000 words
3. Group Project plan: due **16:55, 16th October, 2026**
4. Oral Presentation:
 - File uploaded before **16:55, 1st Nov., 2026 (Sun)**
 - During last 3 tutorial sessions
 - 12 Minutes plus 3 minutes Q&A
5. Essay: Due **16:55, 1st December, 2026**

Group Project Logistics

6. On a book (or other media):

- About Mathematics
- Relevant to **Learning Outcomes**

7. Can use more than one (related) book.

8. Not a simple book report:

- Your Interpretation
- Connection to course/Learning Outcomes

Role of Mathematics in Human Civilization

1. Foundation of science/technology
2. Strong presence in wide range of other areas
3. One of the major threads (together with language, science, and the arts) that form the core of human civilization

Role of Mathematics in Human Civilization

“... Mathematics sets the standard of objective truth for all intellectual endeavours; science and technology bear witness to its practical usefulness. Besides language and music, [mathematics] is one of the primary manifestations of the free creative power of the human mind, and it is the universal organ for world-understanding through theoretical construction. Mathematics must therefore remain an essential element of the knowledge and abilities which we have to teach, of the culture we have to transmit, to the next generation.”

- Hermann Weyl



"Hermann Weyl ETH-Bib Portr 00890" by ETH Zürich - ETH-Bibliothek Zürich, Bildarchiv. Source: Wikipedia

Course Outline

1. Study examples gathered from the long history of humankind, around our daily lives, in diverse areas of human activities, so that we will be able to comprehend how mathematics was, and is being, developed as a work of human endeavor with cultural, intellectual, and social contexts.
2. Investigate the role of mathematics in the development of other areas of our civilization
3. Examine the interplay between mathematics and other cultural pursuits such as
 - philosophy
 - the arts
 - science and technology
 - ...and how they have affected each others' development

Level of Mathematics Required

1. Very minimal!
2. You should have intellectual curiosity and be willing to participate in the logical process of reasoning.
3. You can drive a car without understanding how it works (unless the car breaks – we need mechanics, too!).

Teaching Staff



Dr. Ben KANE

Consultation hours:
Tues. 10-13
Office: Run Run Shaw
Building 411



**Dr. Toby Kwing
To LAI**

Office: Hui Oi Chow
Science Building 223
Meetings: By appointment



**Dr. David Hoi
Fung YU**

Office: Hui Oi Chow
Science Building 226
Meetings: By appointment

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

**Mathematics
and the Arts**

**Mathematics
and Education**

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

Survey Questions

1. With many Common Core options, why did you choose this course and what do you expect to see in this course?



<https://gear.cetl.hku.hk/survey/e9df7d4028864aa2afabb5dbc67265d4>

Course Topics: Numbers

Numbers



Bank Account Information		
Starting Balance	01/12/2014	23000
Deposit	07/12/2014	1100
Withdrawal	13/12/2014	1300
New Balance	01/01/2015	22800

What about “-1300”?

What does “3” mean?

Course Topics: Numbers

Number Systems of different cultures

	1	2	3	4	5	6	7	8	9
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HUNDREDS	𐍪𐍪𐍪	𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪	𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪𐍪
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TENS OF THOUSANDS	𐍪𐍪𐍪𐍪𐍪								
HUNDREDS OF THOUSANDS	𐍪𐍪𐍪𐍪𐍪𐍪								

Egyptian hieratic numerals of the New Kingdom (1600–1000bc) used more symbols than before, making numbers more compact but harder to learn to use.

0	〇	零			
1	一	壹	10	十	拾
2	二	貳	20	廿/廿	貳拾
3	三	叁	30	卅	參拾
4	四	肆	40	卌	肆拾
5	五	伍			
6	六	陸	10 ²	百	佰
7	七	柒	10 ³	千	仟
8	八	捌	10 ⁴	万	萬
9	九	玖	10 ⁸	亿	億

©G. Sironi, www.archimedes-lab.org

1 I	21 XXI	41 XLI	61 LXI	81 LXXXI
2 II	22 XXII	42 XLII	62 LXII	82 LXXXII
3 III	23 XXIII	43 XLIII	63 LXIII	83 LXXXIII
4 IV	24 XXIV	44 XLIV	64 LXIV	84 LXXXIV
5 V	25 XXV	45 XLV	65 LXV	85 LXXXV
6 VI	26 XXVI	46 XLVI	66 LXVI	86 LXXXVI
7 VII	27 XXVII	47 XLVII	67 LXVII	87 LXXXVII
8 VIII	28 XXVIII	48 XLVIII	68 LXVIII	88 LXXXVIII
9 IX	29 XXIX	49 XLIX	69 LXIX	89 LXXXIX
10 X	30 XXX	50 L	70 LXX	90 XC
11 XI	31 XXXI	51 LI	71 LXXI	91 XCI
12 XII	32 XXXII	52 LII	72 LXXII	92 XCII
13 XIII	33 XXXIII	53 LIII	73 LXXIII	93 XCIII
14 XIV	34 XXXIV	54 LIV	74 LXXIV	94 XCIV
15 XV	35 XXXV	55 LV	75 LXXV	95 XCV
16 XVI	36 XXXVI	56 LVI	76 LXXVI	96 XCVI
17 XVII	37 XXXVII	57 LVII	77 LXXVII	97 XCVII
18 XVIII	38 XXXVIII	58 LVIII	78 LXXVIII	98 XCVIII
19 XIX	39 XXXIX	59 LIX	79 LXXIX	99 XCIX
20 XX	40 XL	60 LX	80 LXXX	100 C

200 CC 300 CCC 400 CD 500 D 600 DC 700 DCC 800 DCCC 900 CM 1000 M

Possible Progression									
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

www.3dCart.com

Course Topics: Numbers

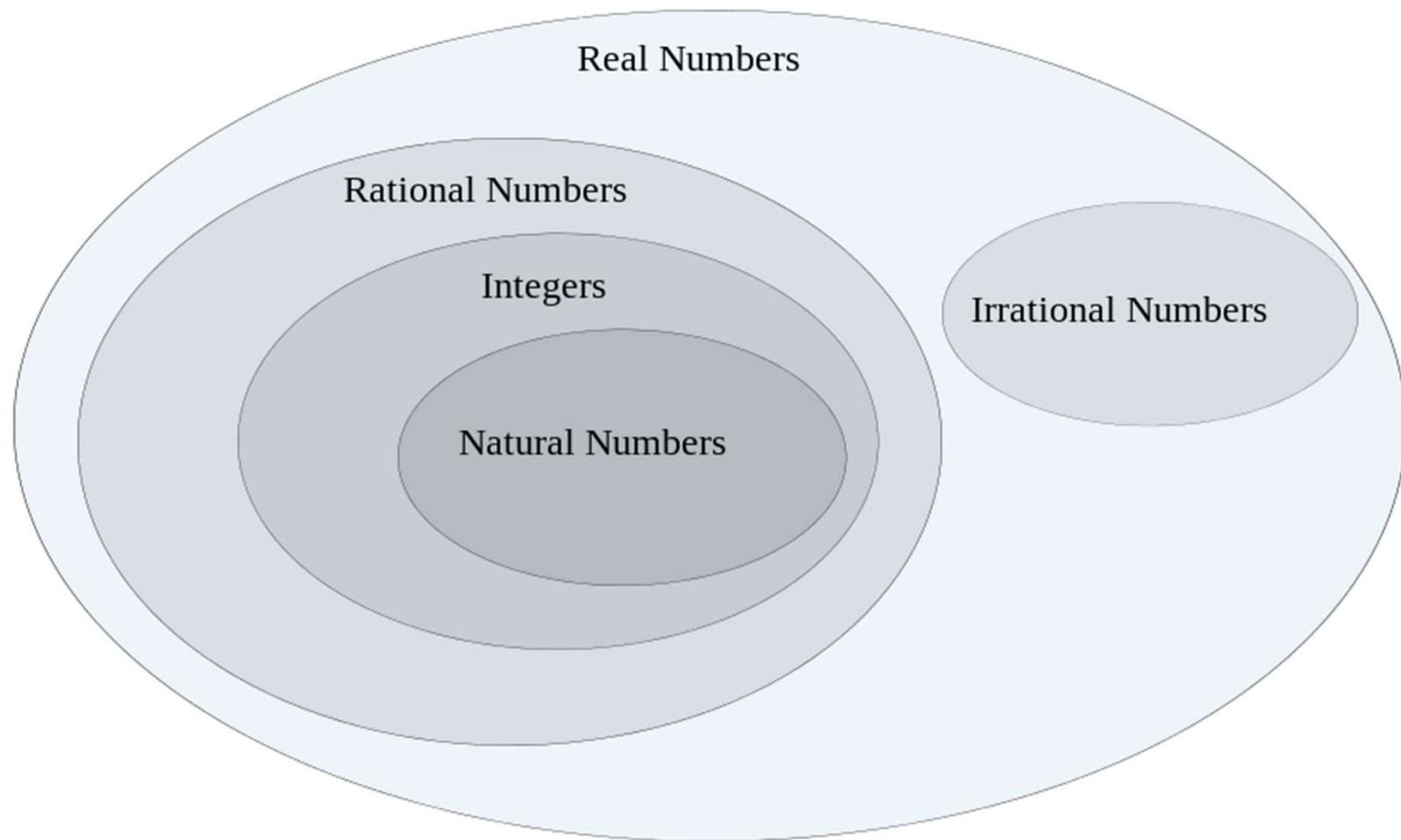
Different kinds of numbers

- Whole/natural numbers: **1, 2, 3, ...**
- Fractions/rational numbers: $\frac{1}{2}, \frac{4}{9}, \frac{11}{13}, \dots$
- Irrational numbers: $\sqrt{7}, \pi \approx 3.14159 \dots$
- Negative numbers: **-3, -7, -10, ...**
- The number **0**
- Imaginary numbers: $i = \sqrt{-1}$

Course Topics: Numbers

- Complex Numbers: $3 + 2\sqrt{-1}, 7 - i$
- Matrices: $\begin{bmatrix} 3 & 7 \\ 4 & 2 \end{bmatrix}, \begin{bmatrix} \frac{5}{3} & \frac{2}{7} \\ 4 & 0 \end{bmatrix}$
- Number fields, Quaternions: ???
- Infinity: ∞

Course Topics: Numbers



"Real numbers" by Damien Karras - Own work.
Licensed under Public Domain via Wikimedia Commons

Course Topics: Math and Education

Mathematics and Education

- What was taught?
- What was the purpose?
- How was it taught?
- Who were taught?
- Who did the teaching?
- Is this the picture you have of Math Education?



Blue Plover: Own Work, © BY-SA 3.0
Creative Commons

Course Topics: Math and Education

Mathematics and Education

- Should we really spend so much time learning mathematics?



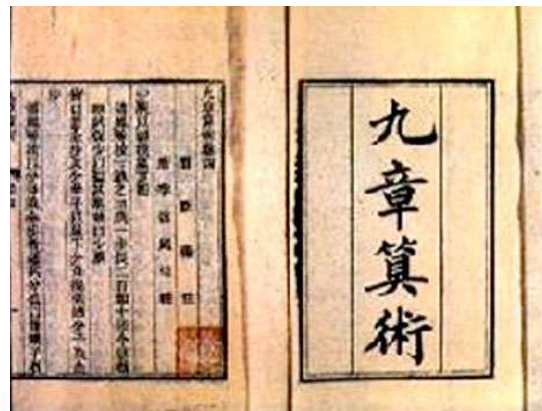
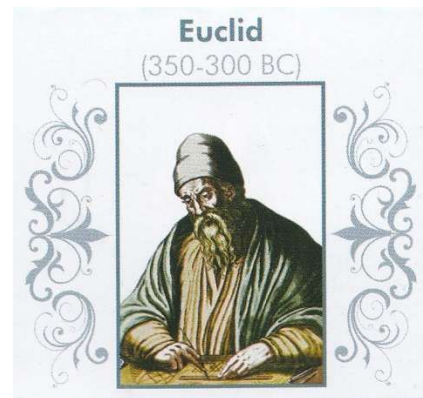
Course Topics: Math and Education

What is the **goal of
Education?**

Course Topics: History of Mathematics

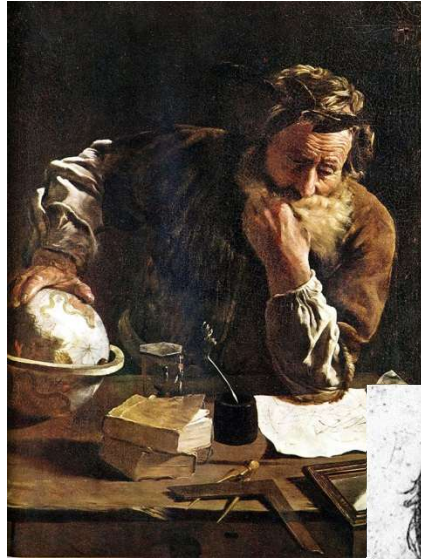
Mathematics in different cultures

- Greek mathematics
- Egyptian mathematics
- Chinese mathematics
- Babylonian mathematics
- ...



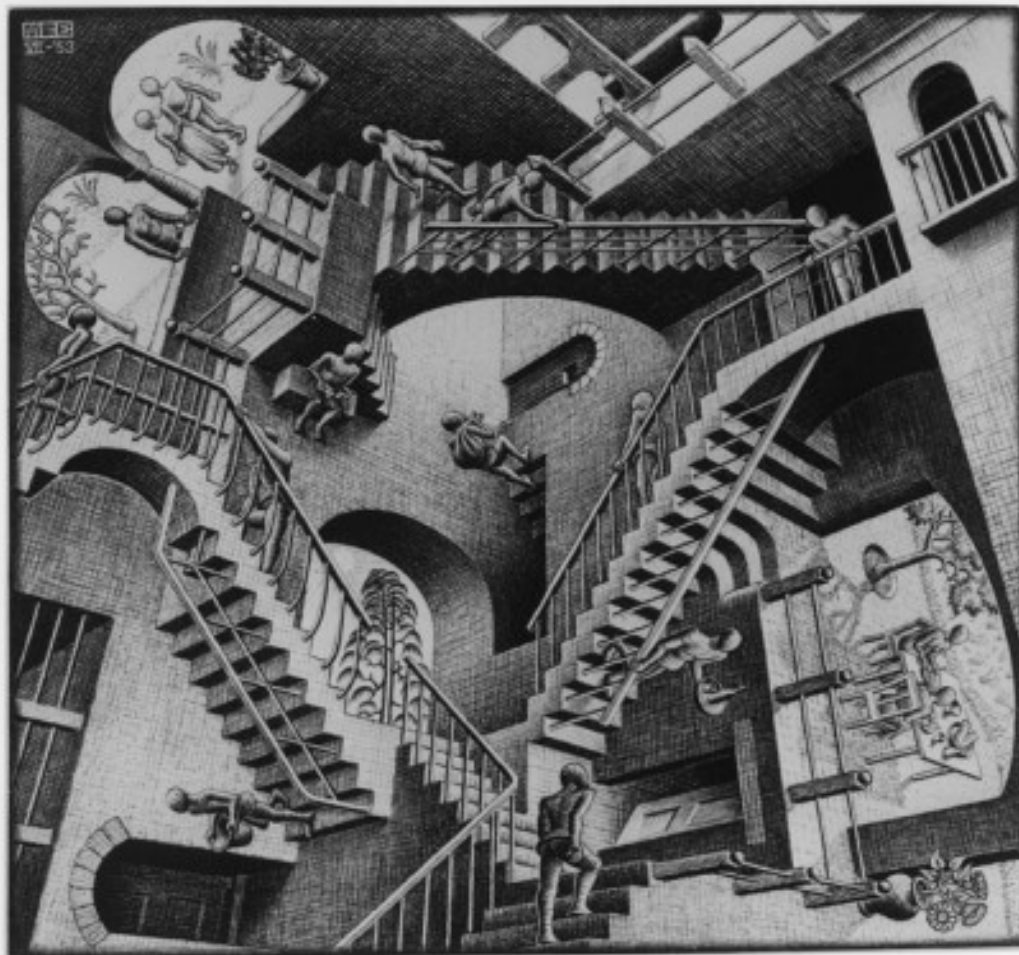
Course Topics: History of Mathematics

A Few Mathematical Lives



Course Topics: Mathematics and the Arts

Mathematics in Art

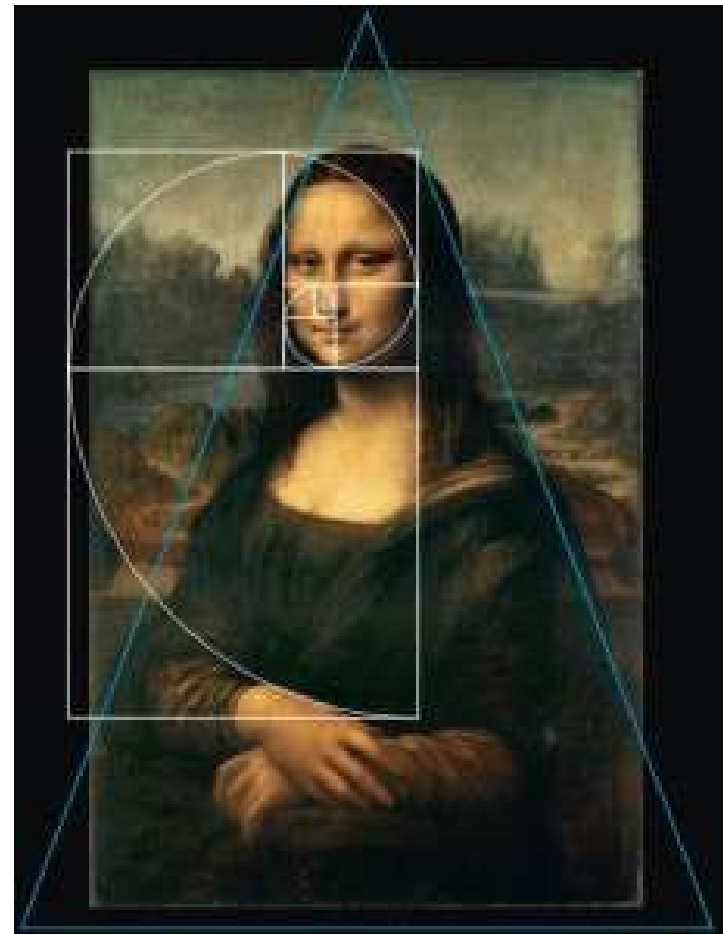
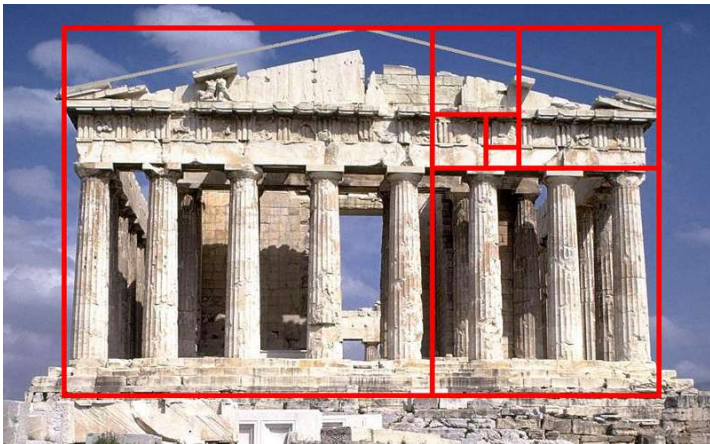


Course Topics:

Mathematics and the Arts

The Golden Ratio $\phi \approx 1.618$

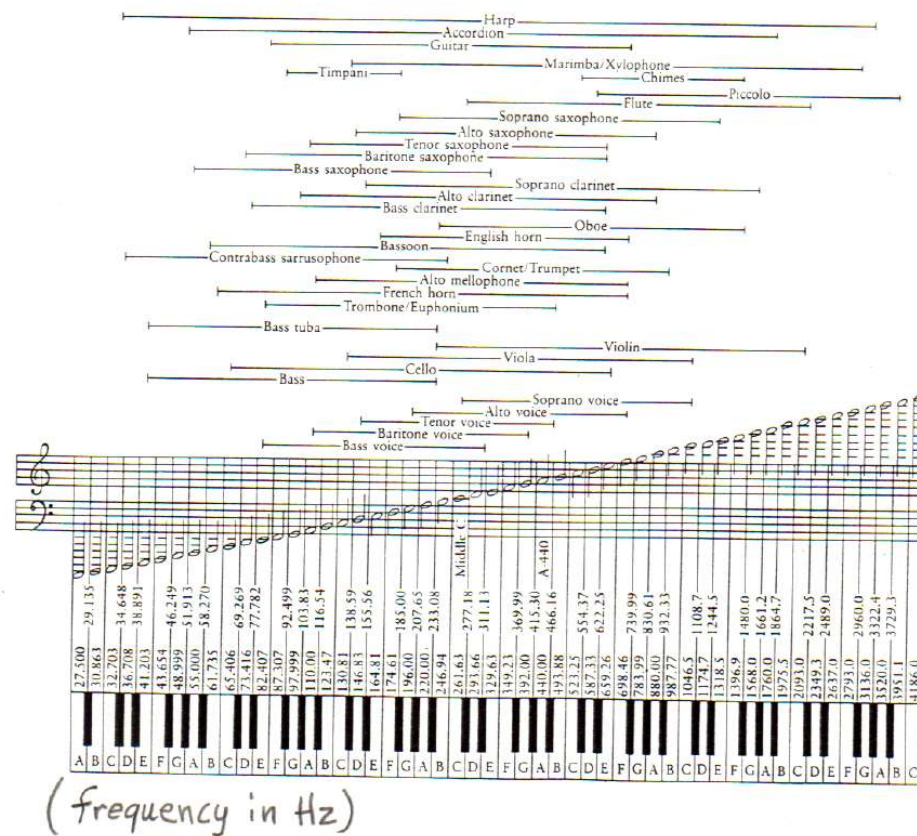
- Appears in Geometry
- Most “eye-pleasing”?
- Used to construct pyramids, parthenon?



Course Topics: Mathematics and the Arts

Mathematics and Music

Figure 8 (Pierce 21)

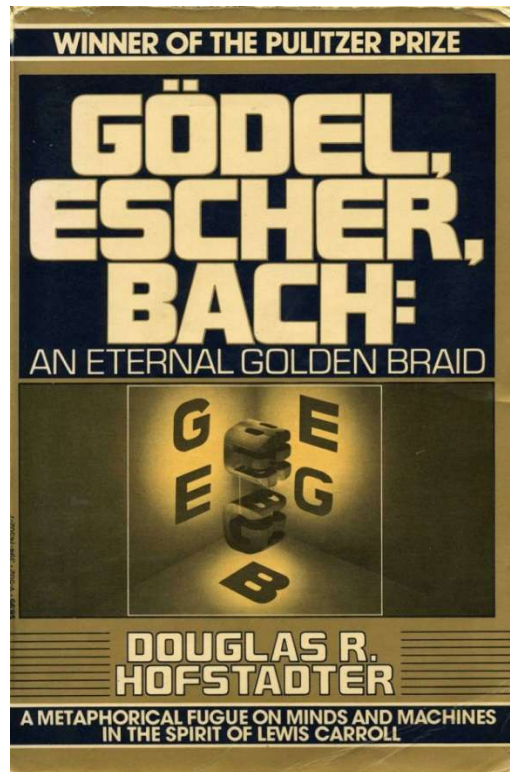


Course Topics: Mathematics and the Arts

Mathematics

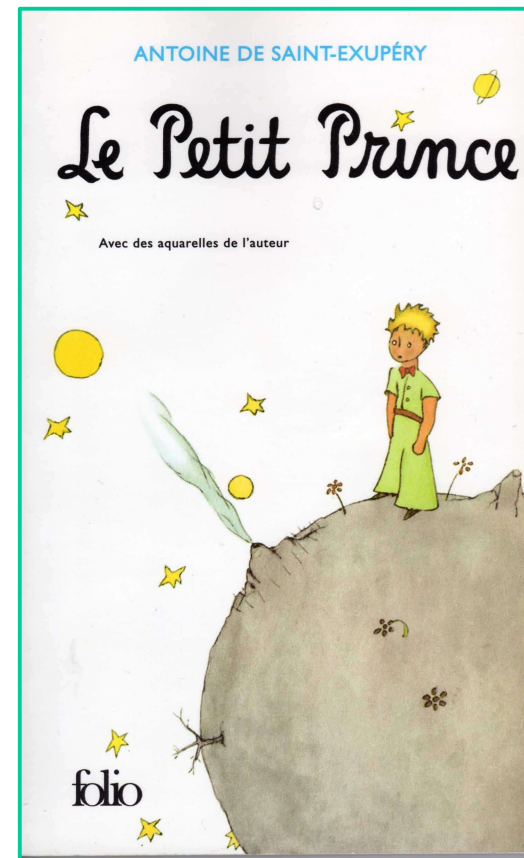
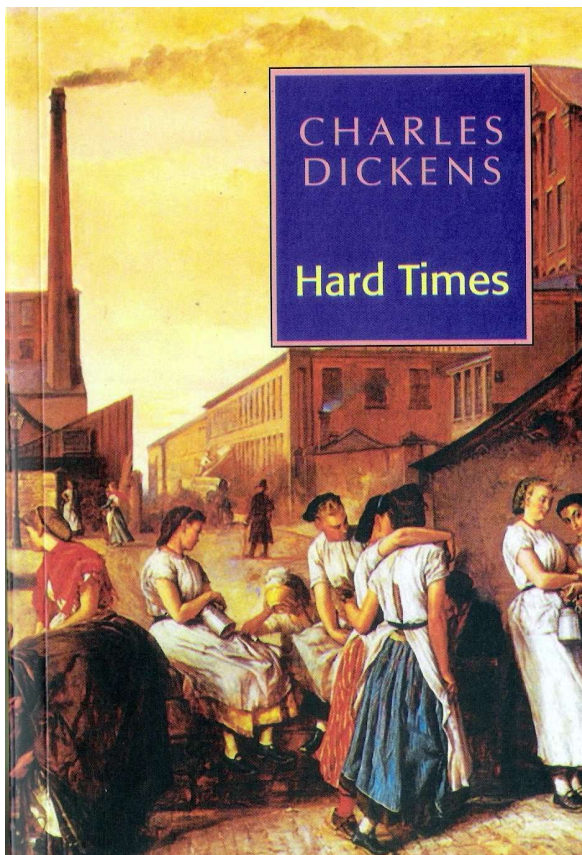
Visual Art

Classical



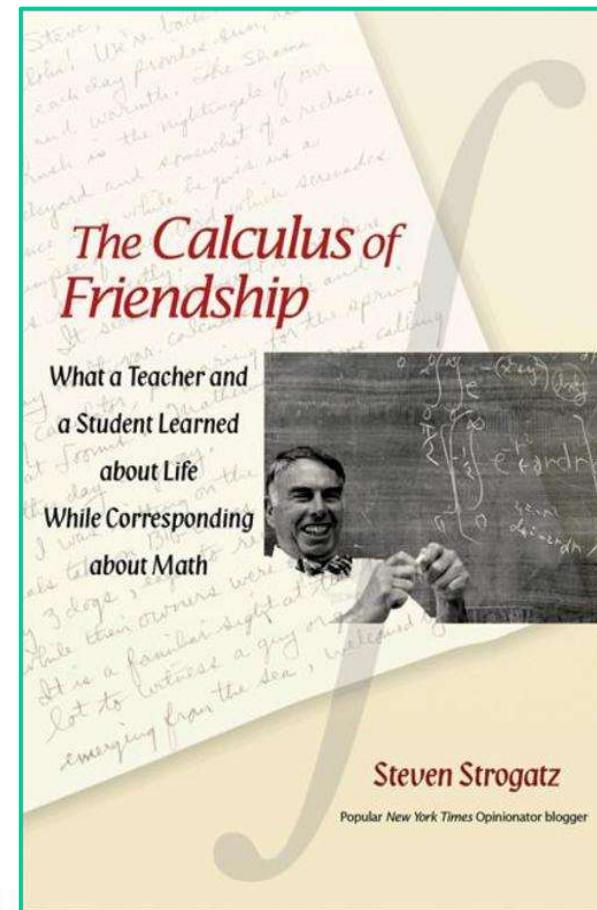
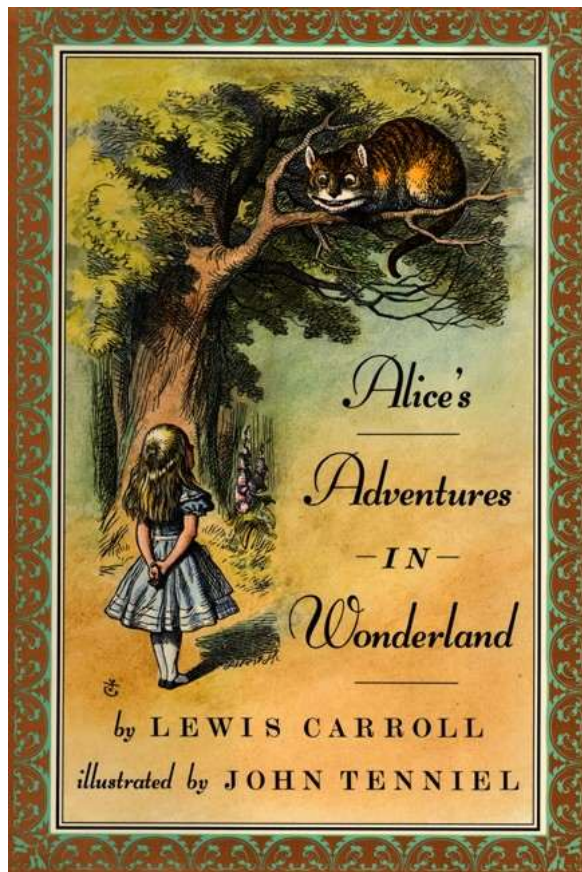
Course Topics: Mathematics, Literature, and TV

Mathematics and Literature



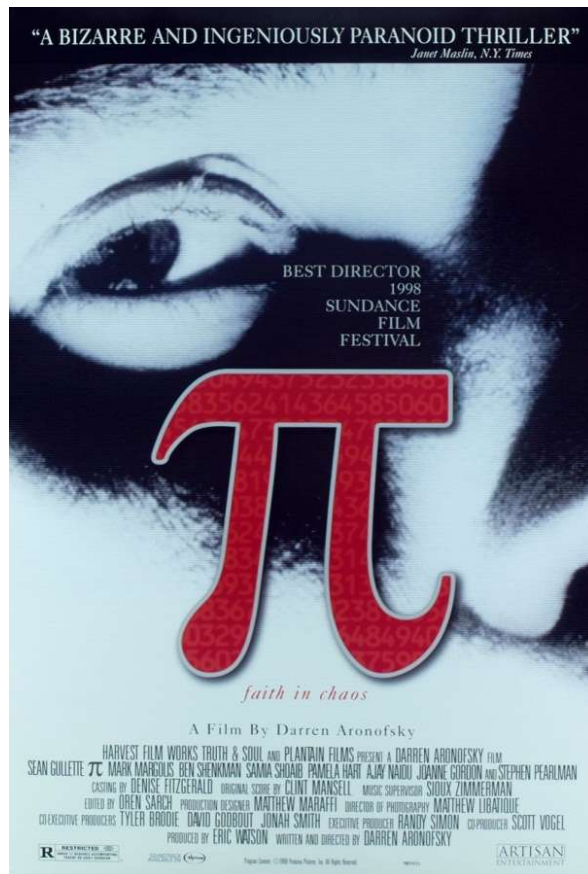
Course Topics: Mathematics, Literature, and TV

Mathematics and Literature



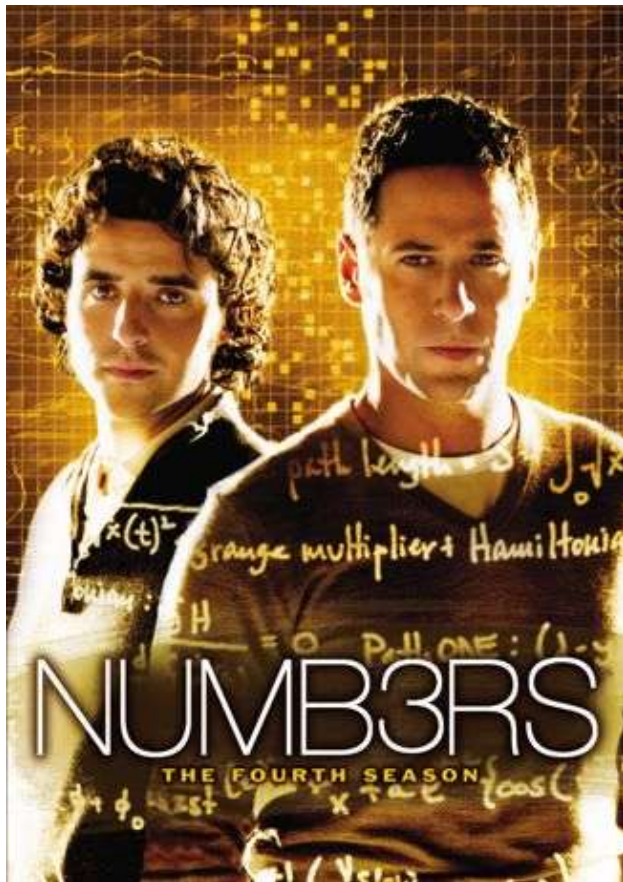
Course Topics: Mathematics, Literature, and TV

Mathematics and Movies



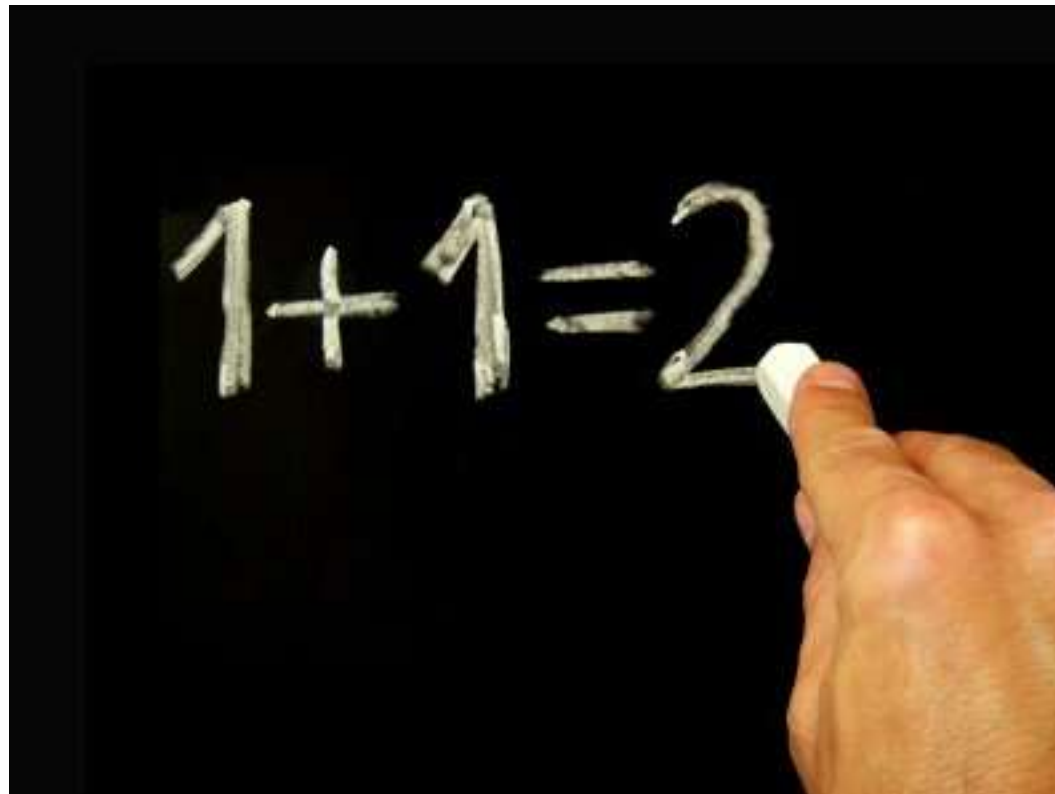
Course Topics: Mathematics, Literature, and TV

Mathematics and TV



Course Topics: Mathematics and Truth

Mathematics and Truth



Course Topics: Mathematics and Truth

Mathematics and Truth

Euclidean Geometry

VS

Non-Euclidean Geometry

Course Topics:

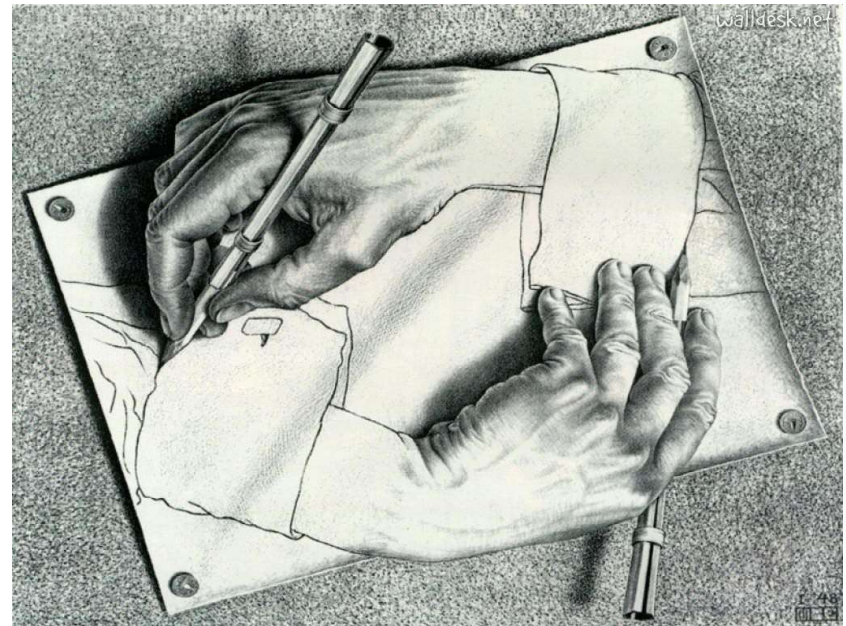
Mathematics and Truth

Mathematics and Truth

- Loops of self-reference



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M. C. Escher, *Drawing Hands*, 1948

Course Topics:

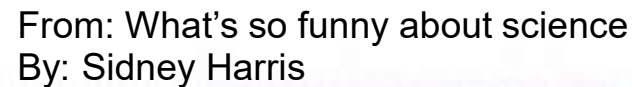
Mathematics and Truth

Mathematics and Truth

- Frege('s), Russell('s), and Hilbert's dream(s)
- Gödel's work and the end of a dream



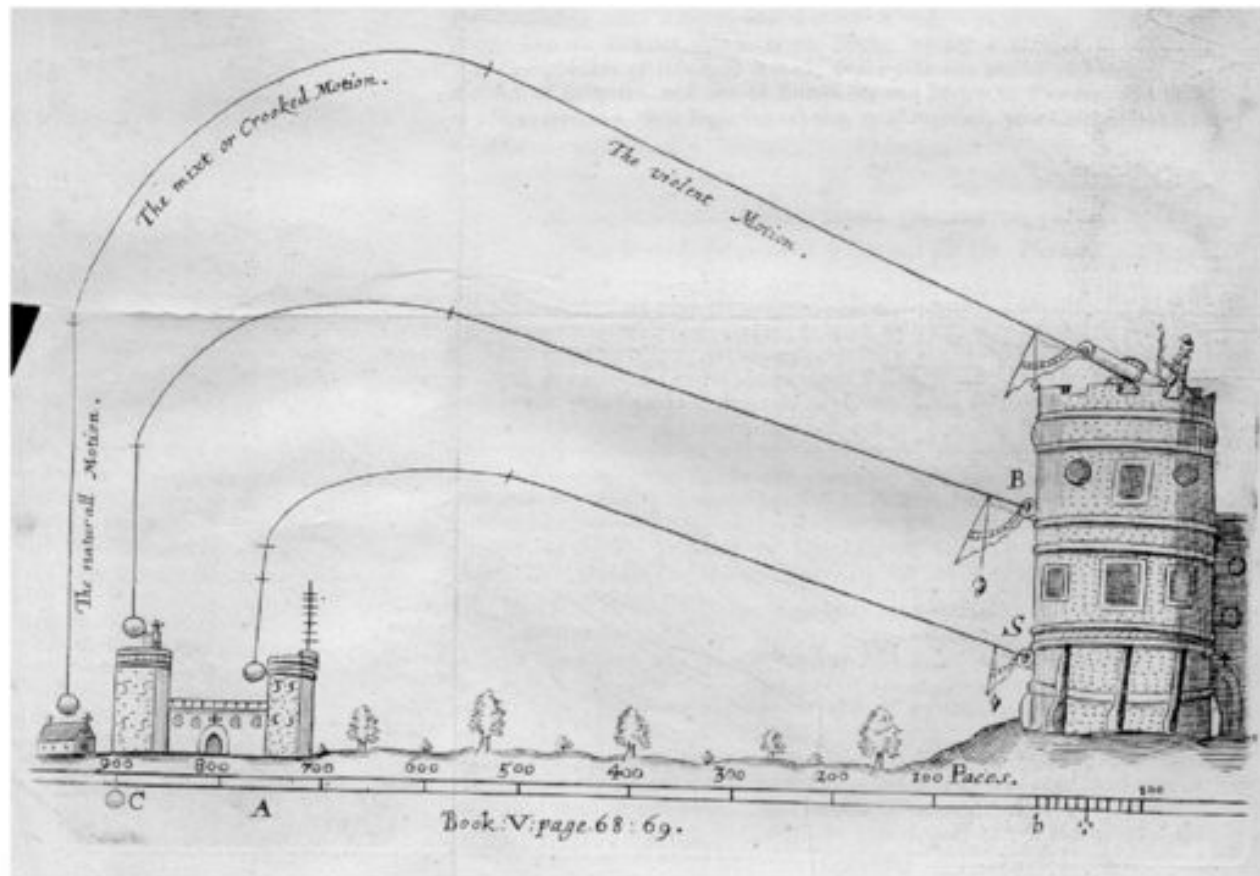
Mathematical Theorems and their Proofs



Course Topics:

Mathematics and Society

Mathematics and Society



Course Topics: **Mathematics and Society**

Mathematics and Society



Course Topics: Mathematics and Society

Mathematics and Society

- Fair social choice/elections??
- Gender bias in Mathematics??



"Nathan Road Police force protesters to South 20141017" by Wing1990hk - Own work. CC BY 3.0, Wikimedia Commons

Course Topics:

Mathematics and Society

Obsession with Numbers

Programme for International Student Assessment (2009) ^{[10][11]} (OECD members as of the time of the study in boldface)											
Maths				Sciences				Reading			
1		Shanghai, China	600	1		Shanghai, China	575	1		Shanghai, China	556
2		Singapore	562	2		Finland	554	2		South Korea	539
3		Hong Kong, China	555	3		Hong Kong, China	549	3		Finland	536
4		South Korea	546	4		Singapore	542	4		Hong Kong, China	533
5		Taiwan	543	5		Japan	539	5		Singapore	526
6		Finland	541	6		South Korea	538	6		Canada	524
7		Liechtenstein	536	7		New Zealand	532	7		New Zealand	521
8		Switzerland	534	8		Canada	529	8		Japan	520
9		Japan	529	9		Estonia	528	9		Australia	515
10		Canada	527	10		Australia	527	10		Netherlands	508
11		Netherlands	526	11		Netherlands	522	11		Belgium	506
12		Macau, China	525	12		Liechtenstein	520	12		Norway	503
13		New Zealand	519	13		Germany	520	13		Estonia	501

Reading Assignment

- Read article:
“Numbers: Easy as 1, 2, 3 – people come into the world ready to count its wonders”, in The Economist, Dec 30, 2008
- URL: www.economist.com/node/12847128
- (may require registration with The Economist, which is free)